Deep Joint Estimation Network for Satellite Video Super-Resolution With Multiple Degradations

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***Abstract*— Super-resolution (SR) for satellite video data has been a hot research topic in the field of remote sensing video analysis. The existing satellite video SR methods assume that the blur kernel in the imaging degradation model is known. However, the blur kernel in real satellite videos is usually unknown, which inevitably results in poor performance when the true blur kernel is not consistent with a predefined blur kernel. To address this issue, this article proposes a deep joint estimation network for satellite video SR (JENSVSR), which jointly estimates blur kernels and SR frames. Specifically, JENSVSR is composed of a video SR subnetwork and a blur kernel estimation subnetwork. On one hand, the video SR subnetwork makes use of multiple video frames to generate super-resolved satellite frames. To effectively fuse information from adjacent frames, an alignment and fusion module is proposed in the feature space. On the other hand, the blur estimation subnetwork is also proposed to predict blur kernels. The two subnetworks are coupled by cross-task feature fusion modules (CTFFMs) to achieve joint estimation rather than two-step independent estimation. The performance of our proposed method is evaluated on synthetic and real satellite videos. The experimental results show that our proposed method is superior to the current state-of-the-art SR methods.**

***Index Terms*— Blur kernel estimation, multiple degradations, satellite video, super-resolution (SR).**

I. Introduction

With the development of remote sensing techniques, several satellites can capture remote sensing videos at a high frame rate, such as the SkySat series [1] and Jilin-1 series. With the characteristics of obtaining large-scale and dynamic scenes, satellite videos provide a new supervised means for applications that demand rapid temporal changes, e.g., disaster monitoring. However, the spatial resolution of satellite videos is about 1 m ground sampling distance (GSD), which is much lower than the spatial resolution of traditional surveillance videos on the ground and unmanned aerial vehicle (UAV) videos. In order to enhance objects in satellite videos, it is necessary to super-resolve satellite videos into higher spatial resolution, which is beneficial to subsequent processing, e.g., detection [2], [3] and tracking tasks [1]. However, SR is still considered as a severely ill-posed inverse problem since low-resolution (LR) images loss information compared with high-resolution (HR) images, especially loss of high-frequency components.

Recently, learning-based methods have been blooming with a more powerful SR capability, especially convolutional neural network (CNN)-based methods. These methods generally need training samples to learn a mapping function between LR–HR image pairs. Most of these methods [4]–[11] are specialized for a single degradation (e.g., bicubic downsampling, or downsampling with a fixed blur kernel). However, the degradation process is much more complex in practical applications, and deviations often exist between the assumed degradation process and the true one. These deviations severely deteriorate SR performance in practical scenarios [12]. Therefore, this kind of methods lacks the flexibility to deal with multiple degradations using a single model.

More attention has been gradually paid to blind SR methods based on deep learning, which super-resolve LR images/videos with unknown degradations, e.g., unknown blur kernels or unknown blur kernels and noise levels. This article only focuses the case of unknown blur kernels, which plays a more important role in SR [12]. Blind SR methods can be roughly divided into two categories: the two-step scheme [13]–[16] and the one-step scheme [17]–[20]. In the two-step scheme, blur kernels are first estimated by one model, and then the other model super-resolves LR images based on estimated blur kernels. In terms of two independent models, SR performance of this scheme mainly relies on the estimated accuracy of blur kernels. It will lead to performance drop when estimated blur kernels are mismatched with true ones and the two models are not well-compatible to each other. The one-step scheme simultaneously estimates blur kernels and SR results with dependent models, including the iterative framework [17], [18] and the non-iterative framework [19], [20]. The iterative framework alternately estimates blur kernels and reconstructs SR results. The iterative framework builds effective feedbacks between two tasks, making more tolerant to estimation errors. But it needs more inference time and it needs to choose the number of iterations. Similar to the two-step scheme, the non-iterative framework estimates blur kernels and reconstructs SR results in a sequential way. The main difference is that this framework is trained in an end-to-end way. However, it only uses LR images to estimate blur kernels, making it difficult to accurately estimate blur kernels.

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To address the problems mentioned above, it is natural to research how to make use of HR information when estimating blur kernels in the one-step scheme with the non-iterative framework. In addition, to the best of our knowledge, there is still lack of work to handle with unknown blur kernels for satellite video super-resolution (SVSR). To achieve the above aim, in this article, we propose a joint estimation network for SVSR (JENSVSR), which couples blur kernel estimation and SR in a parallel way instead of the traditional sequential form in [17]–[19]. The JENSVSR framework consists of two subnetworks. On one hand, an SVSR network (SVSRNet) is constructed to use temporal information of satellite videos, which is composed of a shallow feature extraction layer, a temporal alignment and fusion module, and a reconstruction module. The temporal alignment and fusion module first aligns and fuses multiple adjacent frames in the feature space. Then the reconstruction module extracts spatial features, restores high-frequency information, and maps HR features to the HR frame with sub-pixel convolution layers. On the other hand, the blur kernel estimation network (BKENet) is designed for predicting the blur kernel for the center frame in the input LR sequence. Note that the features in two subnetworks are integrated by developed cross-task feature fusion modules (CTFFMs) to achieve joint estimation. Finally, super-resolved results in case of multiple blur kernels can be obtained via a JENSVSR model.

The main contributions of our work are threefolds.

1) We propose a joint estimation network for SVSR by coupling the estimation of blur kernel and SR via a parallel way, which is composed of SVSRNet and BKENet. The SVSRNet mines the spatio-temporal information to reconstruct SR videos. BKENet predicts blur kernel. Hence, the proposed model can achieve better SR performance in the case of multiple degradations with unknown blur kernels.
2) A module of alignment and fusion is proposed to fuse multiple adjacent frames in the feature space. In this module, feature patches in the center frame search the most similar feature patches from adjacent feature frames to align adjacent features with the center features. Then aligned features are fused by several convolutional layers.
3) We propose a CTFFM that is used to integrate intermediate features between SVSRNet and BKENet. A joint loss function that is made up of SR loss and blur estimation loss is used to train the JENSVSR.

The remainder of this article is organized as follows. In Section II, related works are reviewed. In Section III, the proposed JENSVSR framework is described in detail. The experimental results on the Jilin-1 satellite video datasets are given in Section IV. Finally, the work of this article is summarized in Section V.

II. Related Works

A. SR Based on CNN With a Single Degradation

In terms of simplified assumption with a single degradation, current studies mainly focus on how to design reasonable networks to enhance the capability to learn the mapping function between LR images/videos and HR images/videos. For example, to extract rich image features, many CNN-based SR methods have been developed by designing network architectures, such as Laplacian pyramid networks [21], recurrent networks [22], much wider and deeper networks [23], [24], and attention networks [8]. Most of these methods show significant improvements and have stronger capabilities to learn mapping functions. However, they reconstruct limited high-frequency components due to the complexity in practical scenarios. To solve this problem, prior knowledge of images is also taken into consideration in some CNN-based SR methods, such as generative adversarial networks (GANs) of SR [25] and edge-guided SR networks [26]. Although CNN-based image SR methods can also be used for video SR, they are not very effective due to the ignorance of temporal complementary information. Most CNN-based video SR methods embed the temporal dependency by aligning multiple frames. According to the ways of temporal alignment, video SR methods can be roughly divided into SR with explicit motion estimation and implicit motion estimation. Video SR methods with explicit motion estimation apply optical flows for alignment, such as video ESPCN (VESPCN) [27], recurrent back-projection network (RBPN) [10], and frame recurrent video SR (FRVSR) [28]. Although they have achieved good performance, they are easily impacted by the accuracy of optical flow estimation. Video SR methods with implicit motion estimation focus on designing reasonable fusion modules to make use of temporal complementary information, including 3-D convolutions [29], deformable convolutions [9], [30], and recurrent structures [31].

B. SR Based on CNN With Multiple Degradations

To handle various degradations in real-world image SR, both a set of degradations (e.g., different blur kernels) and LR images are taken as inputs during the training stage in [13]–[16]. These non-blind methods need to know true degradations at the inference stage. Since SR and estimation of degradations are independent of each other, SR performance highly relies on the estimation accuracy of degradation. To address this problem, Gu *et al*. [17] propose an iterative kernel correction (IKC) method, which iteratively estimates and corrects the blur kernel when reconstructing the super-resolved image. Luo *et al*. [18] propose a deep alternating network (DAN), which alternatively estimates the blur kernel by an estimator and restores the SR image by a restorer. Both the IKC and DAN methods achieve coupling between degradation estimation and SR by iterative feedbacks. They also need iterative estimation in the inference stage. Unlike these two methods, Wang *et al*. [19] propose a degradation-aware SR (DASR) network, which learns degradation representations by contrastive learning instead of explicitly estimating degradations. Kim *et al*. [20] propose a kernel-oriented local adjustment network (KOALAnet), which maps the estimated blur kernel as local filtering operations to modulate SR features. Pan *et al*. [32] propose to simultaneously estimate blur kernels and restore HR videos, where a deconvolution method is embedded to generate sharper features with estimated blur

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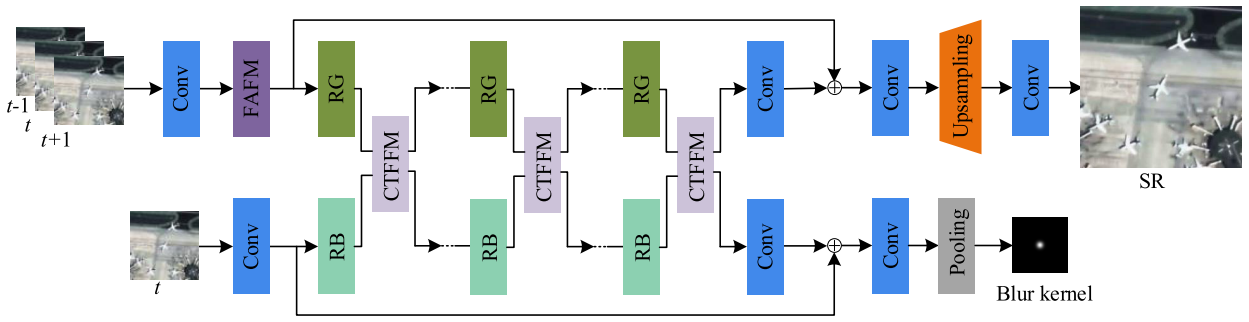


Fig. 1. Overall framework of our proposed JENSVSR. For simplicity, three consecutive LR frames are taken as an example. The upper branch takes the LR temporal sequence as the input and reconstructs the center SR frame, denoted as a satellite video SR subnetwork. The lower branch takes the center LR frame as the input and estimates the blur kernel, denoted as a blur kernel estimation subnetwork. The intermediate features from two subnetworks are fused with each other through CTFFM.

kernels as inputs of the SR network. Another kind of blind SR methods is based on GAN, e.g., CinCGAN [33] and KernelGAN [34]. However, they easily generate artifacts in real-world applications.

C. Satellite Video SR

As for SVSR, due to the difficulty in data acquisition, it has only recently received attention and research. The severely degraded video quality and the wide scene characteristic make SVSR challengeable to reconstruct realistic high-frequency details. In early works, VDSR [35] and ESPCN [36] are re-trained for SVSR. Subsequent works take more account of the characteristics of satellite videos. For example, considering that satellite videos are contaminated by noise, to reduce errors caused by imprecisely explicit motion estimation, an implicit motion estimation method by adaptively fusing spatio-temporal information is proposed in [37]. To enhance spatial resolution of airplanes in satellite videos, Chen *et al*. [38] propose a reflective symmetry shape prior. To better reconstruct high-frequency components, a deep distillation recursive network (DDRN) [39] and a GAN-based edge enhancement network [40] are proposed for SVSR, respectively. Although these two methods have improvements for SVSR, only using spatial information restricts their abilities to reconstruct more texture details. To address this issue, temporal information of satellite videos is used by optical flows in [41], 3-D convolutions in [42] and [43], deformable convolutions in [44], and combination of optical flows and deformable convolutions in [45].

Most SVSR methods predefine one fixed blur kernel when learning SVSR networks. However, the blur kernel in satellite videos is generally unknown, which inevitably causes poor performance when the true blur kernel is not consistent with the assumed one in real-world applications. In view of this situation, the present SVSR methods need to be re-trained when dealing with different blur kernels for SR, which lacks flexibility. Thus far, there are still few works aiming at handling multiple degradations (e.g., unknown blur kernels) for SVSR.

III. Proposed Method

According to the imaging degradation model, the relationship between LR and HR frames in satellite videos can be modeled as

$$\mathbf{y}_t = (\mathbf{x}_t \otimes \mathbf{k})\downarrow_s + \varepsilon \tag{1}$$

where $\mathbf{y}_t$ is an observed frame at time t, and $\mathbf{x}_t$ is an HR frame to be reconstructed. $\mathbf{k}$ is the blur kernel, $\downarrow_s$ is a downsampling operation with scale factor s, and ε is the additive noise.

Given an LR frame $\mathbf{y}_t$ and its $2N$ adjacent frames $\{\mathbf{y}_{t-N}, \ldots, \mathbf{y}_{t-1}, \mathbf{y}_{t+1}, \ldots, \mathbf{y}_{t+N}\}$, we define the center frame $\mathbf{y}_t$ as the target frame to be super-resolved. The goal of our network for the SVSR task is to reconstruct an HR frame that approaches $\mathbf{x}_t$ through SVSRNet.

For the single degradation case, the SVSR task can be expressed as

$$\hat{\mathbf{x}}_t = \text{SVSRNet}(\mathbf{y}_{t-N}, \ldots, \mathbf{y}_t, \ldots, \mathbf{y}_{t+N}; \theta) \tag{2}$$

where θ is the parameter of SVSRNet. In this case, the blur kernel $\mathbf{k}$ is fixed with a known parameter. In the training stage, LR–HR training pairs can be generated according to (1) to learn the parameter θ. In the inference stage, SVSRNet takes the LR video sequences as inputs and generates super-resolved video sequences.

The above case can only handle a specified degradation condition with the known blur kernel. Considering the unknown blur kernel in real-world applications, the blur kernel $\mathbf{k}$ is also simultaneously estimated via BKENet to help the reconstruction of SR results in our proposed JENSVSR. Assuming the input sequence $\{\mathbf{y}_{t-N}, \ldots, \mathbf{y}_t, \ldots, \mathbf{y}_{t+N}\}$ shares the degradation process with the same blur kernel. Then the joint estimation of $\hat{\mathbf{x}}_t$ and $\hat{\mathbf{k}}_t$ can be expressed as

$$\begin{cases} \hat{\mathbf{x}}_t = \text{SVSRNet}(\mathbf{y}_{t-N}, \ldots, \mathbf{y}_t, \ldots, \mathbf{y}_{t+N}; \theta) \\ \hat{\mathbf{k}}_t = \text{BKENet}(\mathbf{y}_t; \theta_{\mathbf{k}}) \end{cases} \tag{3}$$

where $\theta_{\mathbf{k}}$ is the parameter of BKENet. To enhance the SR capacity under multiple degradations with different blur kernels, the intermediate features between SVSRNet and BKNet are fused with each other to achieve joint estimation without iterative optimization. The overall framework of JENSVSR is shown in Fig. 1. The upper branch and lower branch are expressed as SVSRNet and BKENet, respectively. In the SVSRNet branch, consecutive LR frames are first fed into a convolutional layer to extract spatial features. Then, a feature alignment and fusion module (FAFM) is applied to implicitly align and fuse adjacent frames with the

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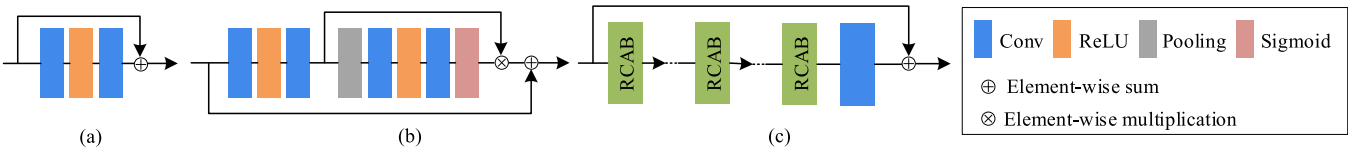


Fig. 2. Architectures of (a) residual block (RB), (b) residual channel attention block (RCAB), and (c) residual group (RG).

center frame. The fused features are then passed through a stack of residual groups (RGs) to further extract SR features. Finally, SR features are further fed in an upsampling layer and a convolutional layer to generate the center SR frame. Meanwhile, the center LR frame is fed into the BKENet branch to generate the estimated blur kernel. BKENet is composed of residual blocks (RBs). In particular, we use the CTFFM to integrate features between SVSRNet and BKENet. It couples two subnetworks and further improves performance of each other under multiple degradations with different blur kernels. In the rest of this section, we describe the architectures of SVSRNet and BKENet and the coupled strategy in detail.

A. SVSR Network

SVSRNet is the primary network for super-resolving satellite videos, which can be roughly divided into two substructures, namely, the FAFM and the reconstruction module. The reconstruction module is built on the work by Zhang *et al.* [8] on an image SR framework with residual channel attention networks (RCANs).

1) Feature Alignment and Fusion Module (FAFM): To make full use of information from adjacent frames in satellite videos to enhance SR performance, it is necessary to fuse adjacent frames with the center frame. The usual solution is to calculate optical flows first, and then warp adjacent frames to the center frame. However, the imaging quality of satellite videos is affected by noise and video compression, and it is not robust to finely align multiple frames with optical flows. Inspired by patch matching for motion estimation, an FAFM is proposed, which is based on patch matching in the feature space to achieve patch-by-patch alignment and temporal information fusion. On one hand, the way of patch-by-patch alignment in the feature space is more robust than using optical flows in the pixel space. On the other hand, the explicit feature alignment by patch matching keeps the clearer physical meaning than the implicit alignment by stacking multiple frames or deformable convolutions. It has more stable and explicit physical constraints.

For convenience, one input neighbor frame is considered here for explanation. Let $\mathbf{F}_t$ denote the center frame feature, and $\mathbf{P}_i^t$ is a local patch centered at position i of $\mathbf{F}_t$. Let the most similar patch in $\mathbf{F}_{t-1}$ to $\mathbf{P}_i^t$ be $\mathbf{P}_j^{t-1}$. Their similarity is calculated by relevance r_{ij} between two patches

$$r_{ij} = \frac{\left\langle \mathrm{vec}\left(\mathbf{P}_i^t\right), \mathrm{vec}\left(\mathbf{P}_j^{t-1}\right)\right\rangle}{\left\|\mathrm{vec}\left(\mathbf{P}_i^t\right)\right\|_2 \left\|\mathrm{vec}\left(\mathbf{P}_j^{t-1}\right)\right\|_2} \tag{4}$$

where $vec(.)$ represents the vectorization operator, and $\langle ., . \rangle$ denotes the inner product.

After searching all the most similar patches for each local patch $\mathbf{P}_i^t$ in $\mathbf{F}_t$, the aligned feature $\mathbf{F}_{t-1\to t}$ can be obtained by averaging all warping local patches $\mathbf{P}_{i\to j}^{t-1}$. All r_{ij} with maximum similarities can form a relevance map $\mathbf{R}_{t-1\to t}$, which can be a regularization to reduce the effect of misalignment because of illumination changes and noise pollution. Then, the aligned feature $\mathbf{F}_{t-1\to t}$ can be further refined as

$$\hat{\mathbf{F}}_{t-1\to t} = \mathbf{F}_{t-1\to t} \odot \mathbf{R}_{t-1\to t} \tag{5}$$

where $\odot$ represents element-wise multiplication.

Similarly, the remaining $2N-1$ adjacent frame features are also processed through the above steps to obtain the corresponding aligned features. Then $2N$ aligned features and the center frame feature are concatenated along the temporal dimension. It can be expressed as

$$\mathbf{F}_t^a = \left[\hat{\mathbf{F}}_{t-N\to t}, \ldots, \hat{\mathbf{F}}_{t-1\to t}, \mathbf{F}_t, \hat{\mathbf{F}}_{t+1\to t}, \ldots, \hat{\mathbf{F}}_{t+N\to t}\right]. \tag{6}$$

The size of $\mathbf{F}_t^a$ is $(2N+1) \times C \times H \times W$, where $(2N+1)$, C, H, and W denote the temporal dimension, feature dimension, height, and width of $\mathbf{F}_t^a$, respectively. Finally, features in $\mathbf{F}_t^a$ are fused by a small network, composed of three convolutional blocks (CBs) and a convolutional layer. The fused network fuses temporal features by gradually squeezing the temporal dimension of $\mathbf{F}_t^a$ to 1. Each CB contains a 3-D convolutional layer and a rectified linear unit (ReLU). Then, the fused features can be represented as

$$\hat{\mathbf{F}} = \mathrm{conv}(\mathrm{CB}(\mathrm{CB}(\mathrm{CB}(\mathbf{F}_t^a)))) \tag{7}$$

where conv is a 2-D convolutional layer with kernel size of 3×3 and stride 1. In CB, the kernel size is $3 \times 3 \times 3$, and the stride is $2 \times 1 \times 1$. The size of $\hat{\mathbf{F}}$ is $1 \times C \times H \times W$.

2) Reconstruction Module: The final fused features output from FAFM are then fed into the following deep reconstruction module. The reconstruction module mainly consists of stacked RGs and two sub-pixel convolutional layers. According to Fig. 2, it can be seen that the basic unit of the RG is the residual channel attention block (RCAB). The channel attention mechanism can model interdependencies across feature channels and adaptively update channel-wise features, which helps the network concentrate on more useful channels. Benefiting from the channel attention mechanism, RGs can strengthen feature extraction and representation to obtain better SR performance. Abundant low-frequency information in the shallow layers can also be passed to deeper layers in RGs with residual connection. It helps the network focus to restore high-frequency information. Finally, the output features of RGs are fed into two sub-pixel convolutional layers and one convolutional layer to generate the super-resolved center frame.

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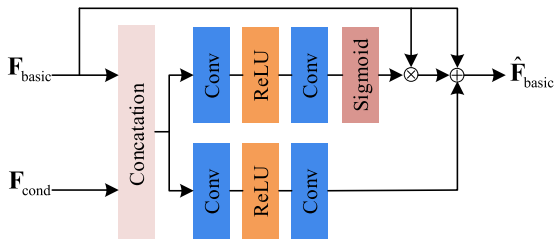


Fig. 3. Architecture of CTFFM.

B. Blur Kernel Estimation Network (BKENet)

Suppose the blur kernel is of size $l \times l$. Following [13], [17], [18], the blur kernel is also first reshaped into l^2 dimension, and then reduced by principal component analysis (PCA) into b dimension. Therefore, the reduced blur kernel $\mathbf{k}$ in low-dimensional space is the variable to be estimated based on BKENet. As shown in Fig. 1, BKENet is composed of one convolutional layer, several RBs, two convolutional layers, and a global average pooling layer. The RBs and convolutional layers estimate the reduced blur kernel $\mathbf{k}$ in a local spatial region. Then, the global average pooling layer obtains the global estimation.

C. Cross-Task Feature Fusion Module

In the proposed JENSVSR framework, we first use SVSRNet and BKENet to extract features from the input LR satellite video sequence, respectively. Let $\mathbf{F}_{\text{SVSRNet}}$ and $\mathbf{F}_{\text{BKENet}}$ denote the features from SVSRNet and BKENet, respectively. Then, the intermediate features are integrated through CTFFM. Let CTFFM_S and CTFFM_B denote CTFFM to generate fused features for SR and blur kernel estimation, respectively. It can be expressed as

$$\begin{cases} \hat{\mathbf{F}}_{\text{SVSRNet}} = \text{CTFFM}_S(\mathbf{F}_{\text{SVSRNet}}, \mathbf{F}_{\text{BKENet}}) \\ \hat{\mathbf{F}}_{\text{BKENet}} = \text{CTFFM}_B(\mathbf{F}_{\text{SVSRNet}}, \mathbf{F}_{\text{BKENet}}). \end{cases} \tag{8}$$

The architecture of CTFFM is shown in Fig. 3. CTFFM is built on the spatial feature transform (SFT) layers [17], [46]. To enhance the feature representation ability for each task, a residual connection is added to the SFT layer. It can be expressed as

$$\hat{\mathbf{F}}_{\text{basic}} = g_1(\mathbf{F}_{\text{concat}}) \odot \mathbf{F}_{\text{basic}} + \mathbf{g}_2(\mathbf{F}_{\text{concat}}) + \mathbf{F}_{\text{basic}} \tag{9}$$

where $\mathbf{F}_{\text{concat}}$ is the concatenated features of $\mathbf{F}_{\text{SVSRNet}}$ and $\mathbf{F}_{\text{BKENet}}$. In CTFFM_S, $\mathbf{F}_{\text{basic}}$ is $\mathbf{F}_{\text{SVSRNet}}$. Similarly, $\mathbf{F}_{\text{basic}}$ is $\mathbf{F}_{\text{BKENet}}$ in CTFFM_B. g_1 and g_2, respectively, represent the scaling and shifting operators in SFT. Instead of the simple concatenation strategy, in CTFFM_S, features of BKENet influence features of SVSRNet through the above affine transformation, which can adaptively modulate features of SVSRNet in transformation. At the same time, features of SVSRNet influence features of BKENet by CTFFM_B. They can be expressed as follows:

$$g_1(\mathbf{F}_{\text{concat}}) = \sigma(\text{conv}(\rho(\text{conv}(\mathbf{F}_{\text{concat}})))) \tag{10}$$

$$g_2(\mathbf{F}_{\text{concat}}) = \text{conv}(\rho(\text{conv}(\mathbf{F}_{\text{concat}}))) \tag{11}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function, and $\rho(\cdot)$ represents the ReLU.

With the help of CTFFM, SVSRNet and BKENet can corporate with each other. Specially, the SVSRNet can make use of information from both LR frames and features of the blur kernel, which makes the SVSRNet better to restore the SR frame with the specified blur kernel. Similarly, BKENet can use information from both LR and SR features, which makes the estimated kernel beneficial to SR.

D. Loss Function

Two loss functions are used to train JENSVSR, including the SR reconstruction loss and the blur kernel estimation loss. The SR reconstruction loss is formulated as

$$L_{\text{SR}} = \|\text{SVSRNet}(\mathbf{y}_{t-N}, \ldots, \mathbf{y}_t, \ldots, \mathbf{y}_{t+N}) - \mathbf{x}_t\|_1. \tag{12}$$

The blur kernel estimated loss is formulated as

$$L_{\text{kernel}} = \|\text{BKENet}(\mathbf{y}_t) - \mathbf{k}\|_1. \tag{13}$$

Finally, the SR loss L_{SR} and the blur kernel estimation loss L_{kernel} are combined as L_{total}

$$L_{\text{total}} = L_{\text{SR}} + L_{\text{kernel}}. \tag{14}$$

With the help of L_{total}, the joint estimation of the JENSVSR framework can achieve end-to-end training.

IV. Experiments

In this section, datasets and experimental details are first introduced. Then, the performance of our proposed method is evaluated on satellite videos.

A. Dataset Setting

In this study, some parts of Jilin-1 satellite videos in [49] are first used to train our proposed JENSVSR, including 6000 video clips with 64 consecutive frames and a spatial size of 256 × 256. Jilin-1 satellite videos are captured by Chang Guang Satellite Technology Company Ltd. The spatial resolution is about 1 m. As shown in Fig. 4, six scene video clips from Jilin-1 satellite videos are used as the test set for synthetic experiments, namely, Jilin-1 satellite video test dataset1. Each scene video clip has 30 consecutive frames with a spatial size of 600 × 600.

In order to further verity our proposed method for real-world experiments without knowing any degradation information, another four Jilin-1 satellite video clips are used to directly input the SR network, namely, Jilin-1 satellite video test dataset2. These scenes are shown in Fig. 5. The spatial size of Scene7 is 285 × 310, and the number of frames is 358. Scene8 has 200 frames and a spatial size of 620 × 674. The spatial sizes of Scene9 and Scene10 are 230 × 280 and 450 × 550, respectively. Scene9 has 300 frames and Scene10 has 358 frames. To further study SR influence when tracking small objects on satellite videos, five small vehicles in these four scenes are chosen as tracked objects as shown in Fig. 5, where the spatial size of these vehicles is less than 20 × 20. After generating SR satellite videos by SR models, super-resolved satellite video clips are input to a tracker to get tracking results. SiamRPN++ [50] as one of the classic single

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(a) (b) (c) (d) (e) (f)

Fig. 4. Some representative frames of the Jilin-1 satellite video test dataset1. (a) Scene1. (b) Scene2. (c) Scene3. (d) Scene4. (e) Scene5. (f) Scene6.



(a) (b) (c) (d)

Fig. 5. Some representative frames of the Jilin-1 satellite video test dataset2. The enlarged areas are five vehicles for object tracking. (a) Scene7. (b) Scene8. (c) Scene9. (d) Scene10.

object tracking methods is chosen as the tracker in this article. In SiamRPN++, the initial coordinates of tracked objects in the first frames are given, and then the predicted coordinates of objects in the following video frames are estimated by SiamRPN++.

B. Implementation Details

In this article, we only focus on SR results for an upscaling factor of 4. For the blur kernels, isotropic Gaussian kernels are only taken into consideration. On one hand, it is because the isotropic Gaussian kernel type is basically consistent with the estimated blur kernels of satellite videos in [37]. One the other hand, this type has been proven practically feasible in image applications [13].

To generate LR and HR training pairs, given a temporal sequence of HR satellite videos, HR frames are first blurred by an isotropic Gaussian kernel of size 21×21 following [17], [19]. The kernel width is randomly sampled from the ranges of $[0.2, 4.0]$. Then the blur frames are downsampled by bicubic downsamplar with a scale factor of 4. Consistent with the settings of [17], [19], the preserving dimension b of the reduced blur kernel is set to 10, where the PCA projection matrix is learned from uniformly sampled blur kernels mentioned above. Our proposed JENSVSR takes seven consecutive frames as inputs. In SVSRNet, the number of RCABs in a RG is set to 20. Both the number of RGs in SVSRNet and RBs in BKENet is set to 10. The Adam method [51] is used for optimization, where $\beta_1 = 0.1$ and $\beta_2 = 0.999$. The initial learning rate is set to 2×10^{-4} and decreased to half after 50 epochs. We train the whole network for 100 epochs with a batch size of 16.

C. Simulated Experiments on Bicubic Degradation

Although our proposed JENSVSR method aims to handle with the SR problem with multiple degradations, where the HR frame is degraded by the unknown blur kernel before downsampling, to validate the effectiveness of SVSRNet in JENSVSR, the first simulated experiments are conducted on in case of a single degradation with bicubic degradation. The LR temporal sequence is generated from the HR sequence of satellite videos by the bicubic downsampling operator. The CTFFM module is not included in SVSRNet. Several representative SR methods are compared, including Bicubic interpolation, ESRGAN [6], EDSR [7], RCAN [8], EDVR [9], RBPN [10], MuCAN [47], VSRT [48], BasicVSR [11], MSDTGP [44], and MSST [45]. ESRGAN, EDSR, and RCAN belong to image SR methods, which process a frame at one time when super-resolving videos. The remaining methods are seven video SR methods. Specially, both MSDTGP and MSST are designed for SVSR. MSST achieves space-time SR for satellite videos, and only spatial SR results are kept here as comparison results. The peak signal-to-noise ratio (PSNR) and the structural similarity index measure (SSIM) are used for quantitative evaluation. PSNR and SSIM are calculated by averaging over all frames for each satellite video scene. The higher the values of PSNR and SSIM, the better the performance of video SR results.

In terms of PSNR and SSIM, the quantitative results of satellite video SR on bicubic degradation are listed in Table I. It can be first seen that bicubic is inferior to other CNN-based methods. The fixed interpolation pattern of bicubic has the limited reconstruction ability compared with the CNN-based methods, which can learn a better mapping function between LR and HR frames with the help of external training datasets. Because of ESRGAN trained by the GAN strategy, it obtains lower PSNR values than other the PSNR-driven SR methods, such as EDSR and RCAN. It can also be seen that the video SR methods (e.g., EDVR, VSRT, MSDTGP, and our proposed SVSRNet) have better SR performance than image SR methods (e.g., RCAN and EDSR) in most video scenes. Specially, according to the average statistics on all

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TABLE I

COMPARISON RESULTS OF PSNR AND SSIM ON SIX SCENES FROM THE JILIN-1 SATELLITE VIDEO TEST DATASET1 BY BICUBIC DOWNSAMPLING DEGRADATION. FOR EACH SCENE, BOTH THE VALUES OF PSNR AND SSIM ARE AVERAGED OVER ALL FRAMES

Methods	Scene1	Scene2	Scene3	Scene4	Scene5	Scene6	Average
Bicubic	28.24 / 0.8766	27.18 / 0.8634	27.25 / 0.8650	32.93 / 0.9221	30.95 / 0.8990	32.59 / 0.8973	29.86 / 0.8872
ESRGAN [6]	30.20 / 0.9139	29.78 / 0.9151	29.59 / 0.9081	35.67 / 0.9489	31.58 / 0.9249	34.25 / 0.9273	31.85 / 0.9230
EDSR [7]	32.13 / 0.9438	31.16 / 0.9365	30.77 / 0.9293	38.05 / 0.9677	34.43 / 0.9522	36.85 / 0.9563	33.90 / 0.9476
RCAN [8]	32.31 / 0.9453	31.37 / 0.9385	31.05 / 0.9328	38.21 / 0.9685	34.70 / 0.9552	37.02 / 0.9579	34.11 / 0.9497
EDVR [9]	31.99 / 0.9451	32.10 / 0.9472	31.38 / 0.9390	38.48 / 0.9702	35.66 / 0.9623	37.09 / 0.9594	34.45 / 0.9539
RBPN [10]	32.29 / 0.9450	32.08 / 0.9461	31.46 / 0.9384	38.38 / 0.9694	35.69 / 0.9616	37.14 / 0.9584	34.51 / 0.9532
MuCAN [47]	32.28 / 0.9456	32.15 / 0.9471	31.54 / 0.9398	38.58 / 0.9704	35.69 / 0.9623	37.26 / 0.9597	34.58 / 0.9542
VSRT [48]	32.35 / 0.9460	32.26 / 0.9481	31.52 / 0.9396	38.54 / **0.9706**	35.85 / 0.9629	37.30 / **0.9602**	34.64 / 0.9546
BasicVSR [11]	32.42 / 0.9464	32.64 / 0.9514	31.48 / 0.9391	38.36 / 0.9695	35.59 / 0.9609	37.23 / 0.9590	34.62 / 0.9544
MSDTGP [44]	32.34 / 0.9459	31.05 / 0.9346	31.44 / 0.9377	37.52 / 0.9644	34.11 / 0.9477	36.53 / 0.9523	33.83 / 0.9471
MSST [45]	29.74 / 0.9193	28.26 / 0.8975	28.89 / 0.9070	32.94 / 0.9419	31.33 / 0.9234	32.67 / 0.9261	30.64 / 0.9192
SVSRNet (Ours)	**32.86** / **0.9512**	**33.34** / **0.9555**	**31.63** / **0.9404**	**38.62** / 0.9698	**36.95** / **0.9668**	**37.34** / **0.9602**	**35.12** / **0.9573**

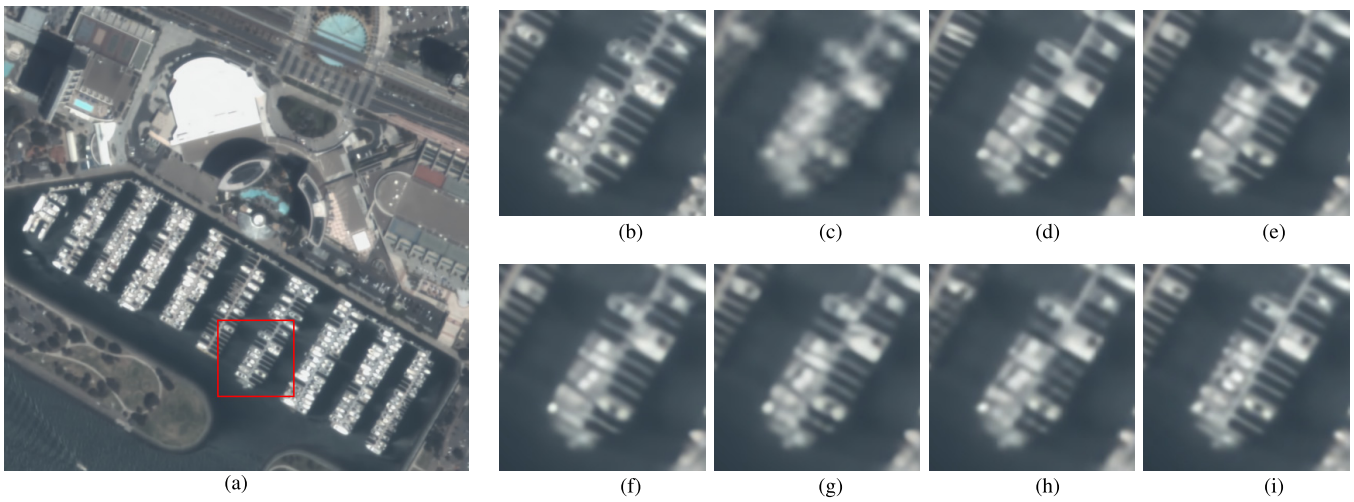


Fig. 6. Qualitative results on the 10th frame of Scene5 with bicubic degradation. The different SR results of the red box area are zoomed in for better visualization. The highest PSNR value is marked in bold. (a) Tenth frame of Scene5. (b) Ground truth. (c) Bicubic (30.98). (d) RCAN (34.70). (e) EDVR (35.88). (f) VSRT (36.06). (g) BasicVSR (35.69). (h) MSDTGP (34.17). (i) Ours (**37.00**).

satellite video scenes, the PSNR and SSIM values of EDVR are 0.34 dB and 0.0042 higher than RCAN, respectively. It is because adjacent frames bring supplementary information, which can provide more description for SR of the center frame. Our proposed SVSRNet achieves the highest PSNR and SSIM values in most satellite video scenes. Especially in Scene5, PSNR of SVSRNet is 1.29 dB higher than EDVR, and 2.84 dB higher than MSDTGP. Considering the average statistics of all the scenes, our SVSRNet leads VSRT by 0.48 dB and leads MSDTGP by 1.29 dB.

We also present some visual results in Figs. 6 and 7. For better comparison of different SR methods, we provide enlarged views in the red box region. It can be first seen that the details of Bicubic SR results are a little over-smoothed, which are not visual-friendly. As shown in Fig. 6, EDVR, BasicVSR, and our SVSRNet basically succeed in reconstructing the ship in the upper left corner, but both VRST and MSDTGP cannot. However, it can also be found that it is more challenging for single image SR methods (e.g., RCAN) to recover objects rightly while keeping sharp edges. With the help of temporal information, video SR methods (e.g., EDVR and BasicVSR) can better handle with the two problems mentioned above. Especially, our proposed SVSRNet can restore more clearer and sharper details than comparison methods. For example, as shown in Fig. 6, the shape of two side-by-side ships in the middle region is successfully recovered by our proposed SVSRNet. Other methods blur the boundary between them. As shown in Fig. 7, in terms of vehicles in the left region, our proposed SVSRNet shows sharper edges. In general, our proposed SVSRNet has superior SR capacity to restore more details and clearer textures.

D. Simulated Experiments on Multiple Degradations

In this subsection, we evaluate the performance of our proposed JENSVSR method on the isotropic Gaussian blur kernel degradation. We compare the proposed JENSVSR method with four blind SR methods without known parameters of blur kernels, including IKC [17], DAN [18], DASR [19]), and DBVSR [32]. Specially, the former three methods (IKC, DAN, and DASR) are developed for blind image SR,

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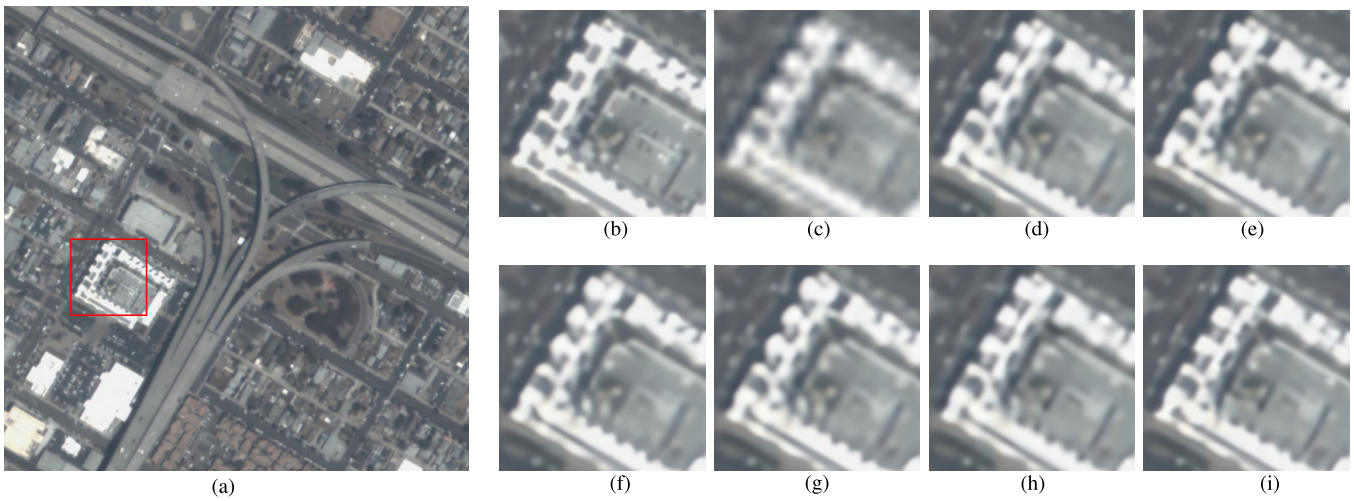


Fig. 7. Qualitative results on the 12th frame of Scene6 with bicubic degradation. The different SR results of the red box area are zoomed in for better visualization. The highest PSNR value is marked in bold. (a) Twelfth frame of Scene6. (b) Ground truth. (c) Bicubic (32.57). (d) RCAN (37.05). (e) EDVR (37.16). (f) VSRT (**37.40**). (g) BasicVSR (37.27). (h) MSDTGP (36.51). (i) Ours (37.34).

and the last one (DBVSR) is proposed for blind video SR. In addition, we also compare SR methods in the two-step scheme, including deblurring before SR and deblurring after SR. PVDnet [52] and MSDTGP [44] are, respectively, selected as the deblurring model and SR model. They are denoted as PVDnet + MSDTGP (also abbreviated PVD + MDP) and MSDTGP + PVDnet (also abbreviated MDP + PVD).

The quantitative results of SR methods with different kernel widths on the Jilin-1 satellite video test dataset1 are shown in Table II. The following observations and analyses can be obtained according to Table II. In terms of PSNR and SSIM, it can be first seen that Bicubic, PVDnet + MSDTGP, and MSDTGP + PVDnet have worse SR performance than other methods in the case of different kernel widths. It is because only the downsampling processing is considered in MSDTGP. In addition, the estimation errors by the deblurring step are accumulated because PVDnet and MSDTGP are two independent models. Second, though DBVSR takes temporal sequences as inputs for blind video SR, it shows lower PSNR and SSIM values than IKC, DAN, and DASR. In DBVSR, the estimated blur kernel by a network is indirectly applied to the SR network using a deconvolution method. The estimated accuracy of blur kernels is implicitly reflected by the SR results. It may bring bigger accumulated errors because of the deconvolution step, compared with IKC, DAN, and DASR. Third, in terms of averaged results on all satellite videos, the SR performance of IKC decreases when increasing the kernel widths. IKC also has less robustness than DAN, DASR, and our proposed JENSVSR method. For example, from kernel width 1.2 to 2.4, the averaged PSNR value of IKC, DAN, DASR, and JENSVSR decreased by 2.39, 0.12, 0.82, and 0.28 dB, respectively. Through the alternative estimation strategy, DAN achieves competitive SR results with the best robustness. Finally, our proposed JENSVSR method yields the highest PSNR and SSIM values in most cases. Especially on Scene3, JENSVSR leads IKC, DAN, and DASR by at least 1.67, 1.34, and 1.39 dB, respectively.

It is noted that both IKC and DAN iteratively estimate the blur kernel and reconstruct the SR frame. Although DASR and JENSVSR do not adopt the iterative strategy, the performance of them is better or competitive compared with IKC and DAN, which indicates that the iteratively alternative optimization is not a necessary step. In terms of SR performance, the cross-task feature fusion strategy in JENSVSR is superior to degradation representation in DASR. It is because both SR and blur kernel estimation tasks are coupled with positive promotion. However, in DASR, the cooperation of degradation representation and SR is limited because of no feedbacks between two tasks in the inference stage.

Some visual results are given in Figs. 8–11. It can be seen that with the increase in kernel widths, the SR results of PVD + MDP are more and more blurred. For the reasons mentioned above, the SR performance of DBVSR decreases quickly when a kernel width is larger than 1.2. In terms of visual performance when degradation by different blur kernels, DAN, DASR, and our proposed JENSVSR show more stable SR results than IKC. When increasing the kernel width of blur kernel, IKC tends to produce over-sharp edges. For example, the SR results of IKC in Figs. 10 and 11 look very unrealistic. Both DAN and DASR also have this tendency, such as the buildings in Fig. 10. Compared with DAN and our proposed JENSVSR, texture details are much unclear in DASR. In general, our proposed JENSVSR can produce more realistic textures, which are faithful to the characteristics of satellite videos. As shown in Fig. 11, our proposed JENSVSR can also restore much clearer edges of two buildings in the upper left corner region than other comparison methods. It can be seen that these edges are totally and partly blurred in other SR methods.

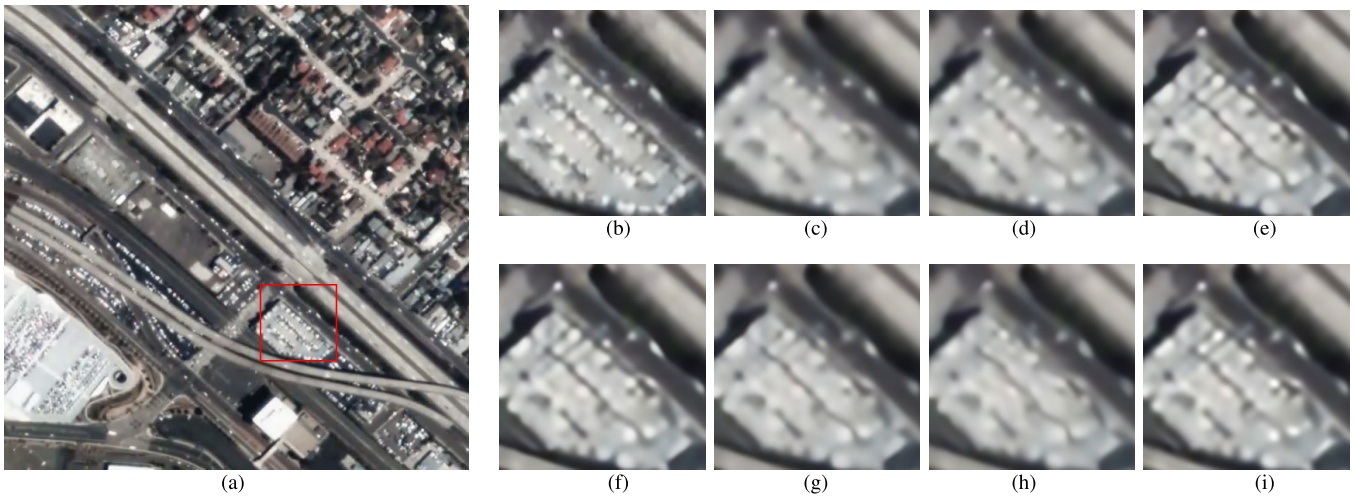
In addition, visualization of estimated blur kernels is shown in Fig. 12, which are estimated on Scene4 with a kernel width of 2.0. Compared with IKC, the estimated blur kernel of JENSVSR shows a much similar shape to the truth map.

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TABLE II

COMPARISON RESULTS OF PSNR AND SSIM ON SIX SCENES FROM THE JILIN-1 SATELLITE VIDEO TEST DATASET1 ON MULTIPLE DEGRADATIONS. FOR EACH SCENE, BOTH THE VALUES OF PSNR AND SSIM ARE AVERAGED OVER ALL FRAMES

Kernel Width	Test Scenes	Bicubic	PVDnet [52]+ MSDTGP [44]	MSDTGP [44]+ PVDnet [52]	IKC [17]	DAN [18]	DASR [19]	DBVSR [32]	JENSVSR (Ours)
1.2	Scene1	26.86 / 0.8432	29.24 / 0.8898	30.52 / 0.9242	31.78 / 0.9406	31.59 / 0.9412	31.13 / 0.9317	29.80 / 0.9123	**32.13** / **0.9424**
	Scene2	26.15 / 0.8347	28.70 / 0.8907	29.69 / 0.9162	**31.17** / **0.9364**	30.38 / 0.9334	30.11 / 0.9234	29.00 / 0.9057	31.16 / 0.9344
	Scene3	25.88 / 0.8313	28.83 / 0.8949	29.91 / 0.9186	29.38 / 0.9193	29.38 / 0.9224	29.66 / 0.9178	28.84 / 0.9009	**31.05** / **0.9316**
	Scene4	31.70 / 0.9056	34.14 / 0.9305	35.69 / 0.9526	**37.53** / **0.9660**	35.95 / 0.9617	35.98 / 0.9574	34.93 / 0.9475	37.34 / 0.9631
	Scene5	29.82 / 0.8785	31.84 / 0.9100	32.90 / 0.9322	33.86 / **0.9484**	32.48 / 0.9437	32.92 / 0.9386	32.36 / 0.9261	**34.10** / 0.9468
	Scene6	31.35 / 0.8733	33.24 / 0.9024	35.11 / 0.9374	35.94 / 0.9516	35.98 / **0.9527**	35.34 / 0.9430	34.27 / 0.9283	**36.20** / 0.9492
	Average	28.63 / 0.8611	31.00 / 0.9030	32.30 / 0.9302	33.28 / 0.9437	32.63 / 0.9425	32.52 / 0.9353	31.53 / 0.9201	**33.66** / **0.9446**
1.6	Scene1	25.96 / 0.8164	27.68 / 0.8562	28.58 / 0.8928	32.05 / 0.9413	31.64 / 0.9412	30.99 / 0.9280	28.07 / 0.8810	**32.17** / **0.9420**
	Scene2	25.46 / 0.8120	27.29 / 0.8600	28.12 / 0.8893	31.13 / 0.9339	30.45 / 0.9335	29.91 / 0.9191	27.54 / 0.8779	**31.21** / **0.9342**
	Scene3	24.97 / 0.8036	27.13 / 0.8612	28.13 / 0.8910	26.97 / 0.8927	29.38 / 0.9223	29.65 / 0.9161	27.20 / 0.8716	**31.11** / **0.9317**
	Scene4	30.88 / 0.8924	32.66 / 0.9138	34.01 / 0.9380	37.31 / **0.9649**	35.93 / 0.9611	36.23 / 0.9569	33.10 / 0.9298	**37.33** / 0.9626
	Scene5	29.06 / 0.8623	30.59 / 0.8892	31.44 / 0.9121	33.85 / **0.9479**	32.50 / 0.9436	33.04 / 0.9378	30.97 / 0.9053	**34.13** / 0.9461
	Scene6	30.54 / 0.8541	31.88 / 0.8774	33.56 / 0.9179	35.70 / 0.9498	**36.09** / **0.9524**	35.47 / 0.9420	32.62 / 0.9042	36.07 / 0.9482
	Average	27.81 / 0.8401	29.54 / 0.8763	30.64 / 0.9068	32.84 / 0.9384	32.67 / 0.9424	32.55 / 0.9333	29.92 / 0.8950	**33.67** / **0.9441**
2.0	Scene1	25.00 / 0.7835	26.19 / 0.8158	26.76 / 0.8496	31.94 / 0.9396	31.74 / **0.9408**	30.61 / 0.9221	26.39 / 0.8383	**32.12** / 0.9404
	Scene2	24.73 / 0.7848	25.96 / 0.8227	26.64 / 0.8538	30.53 / 0.9262	30.54 / **0.9331**	29.64 / 0.9142	26.12 / 0.8401	**31.18** / 0.9328
	Scene3	23.99 / 0.7691	25.47 / 0.8175	26.43 / 0.8536	26.06 / 0.8787	29.76 / 0.9247	29.19 / 0.9113	25.52 / 0.8296	**31.10** / **0.9308**
	Scene4	30.00 / 0.8764	31.24 / 0.8941	32.31 / 0.9175	35.08 / 0.9562	35.57 / 0.9581	36.01 / 0.9543	31.44 / 0.9071	**37.26** / **0.9609**
	Scene5	28.24 / 0.8428	29.34 / 0.8646	29.95 / 0.8854	33.40 / **0.9448**	32.44 / 0.9419	32.78 / 0.9348	29.55 / 0.8786	**34.08** / 0.9440
	Scene6	29.68 / 0.8312	30.61 / 0.8492	31.84 / 0.8883	34.61 / 0.9419	**36.09** / **0.9508**	35.20 / 0.9380	31.05 / 0.8728	35.97 / 0.9461
	Average	26.94 / 0.8146	28.13 / 0.8440	28.99 / 0.8747	31.94 / 0.9312	32.69 / 0.9416	32.24 / 0.9291	28.35 / 0.8611	**33.62** / **0.9425**
2.4	Scene1	24.08 / 0.7471	24.91 / 0.7733	25.23 / 0.8002	30.23 / 0.9252	31.59 / **0.9372**	30.15 / 0.9160	24.97 / 0.7903	**31.84** / 0.9354
	Scene2	24.01 / 0.7552	24.84 / 0.7838	25.38 / 0.8136	29.15 / 0.9203	30.42 / **0.9302**	29.45 / 0.9101	24.91 / 0.7984	**30.99** / 0.9291
	Scene3	23.04 / 0.7301	24.04 / 0.7690	24.86 / 0.8075	26.03 / 0.8771	29.40 / 0.9207	28.14 / 0.9005	24.05 / 0.7801	**30.94** / **0.9278**
	Scene4	29.12 / 0.8589	30.00 / 0.8736	30.69 / 0.8918	33.57 / 0.9467	35.65 / 0.9562	35.47 / 0.9499	30.05 / 0.8826	**36.94** / **0.9573**
	Scene5	27.44 / 0.8213	28.22 / 0.8390	28.61 / 0.8554	32.36 / 0.9367	32.42 / 0.9389	32.25 / 0.9296	28.30 / 0.8493	**33.87** / **0.9396**
	Scene6	28.84 / 0.8062	29.50 / 0.8204	30.12 / 0.8478	33.97 / 0.9349	35.55 / **0.9449**	34.74 / 0.9321	29.72 / 0.8383	**35.72** / 0.9412
	Average	26.09 / 0.7865	26.92 / 0.8098	27.48 / 0.8360	30.89 / 0.9235	32.51 / 0.9380	31.70 / 0.9230	27.00 / 0.8232	**33.38** / **0.9384**



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 8. Qualitative results on the 15th frame of Scene1 with a kernel width of 1.2. The different SR results of the red box area are zoomed in for better visualization. The highest PSNR value is marked in bold. (a) Fifteenth frame of Scene1. (b) Ground truth. (c) PVD + MDP (29.06). (d) MDP + PVD (30.34). (e) IKC (31.62). (f) DAN (31.23). (g) DASR (30.91). (h) DBVSR (29.59). (i) Ours (**31.93**).

JENSVSR has a better capability to simultaneously estimate the blur kernel and super-resolve frames than IKC.

E. Experimental Results of Real-World Satellite Videos

To further evaluate the generalization ability of our proposed JENSVSR method on practical scenarios, satellite video clips on the Jilin-1 data satellite video test dataset2 are directly input to JENSVSR instead of simulating LR videos as inputs. Natural image quality evaluator (NIQE) [53] is used to measure the quality of super-solved frames. NIQE represents the distance between the feature model of natural scene statistic and the feature model extracted from the distorted

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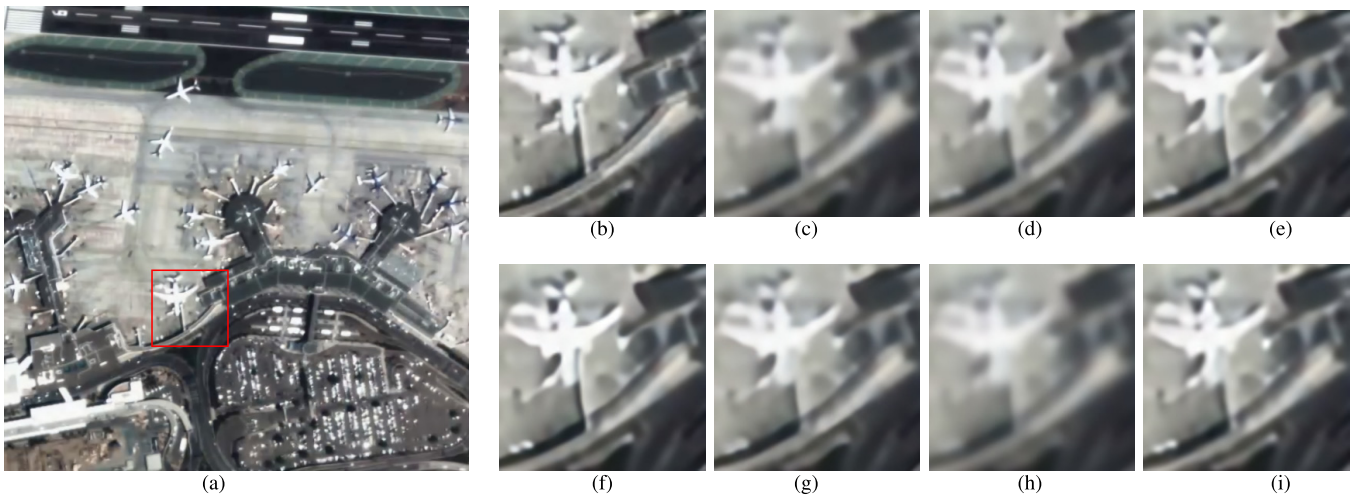


Fig. 9. Qualitative results on the 18th frame of Scene2 with a kernel width of 1.6. The different SR results of the red box area are zoomed in for better visualization. The highest PSNR value is marked in bold. (a) Eighteenth frame of Scene2. (b) Ground truth. (c) PVD + MDP (27.24). (d) PVD + MDP (28.07). (e) IKC (31.09). (f) DAN (30.41). (g) DASR (29.88). (h) DBVSR (27.51). (i) Ours (**31.19**).

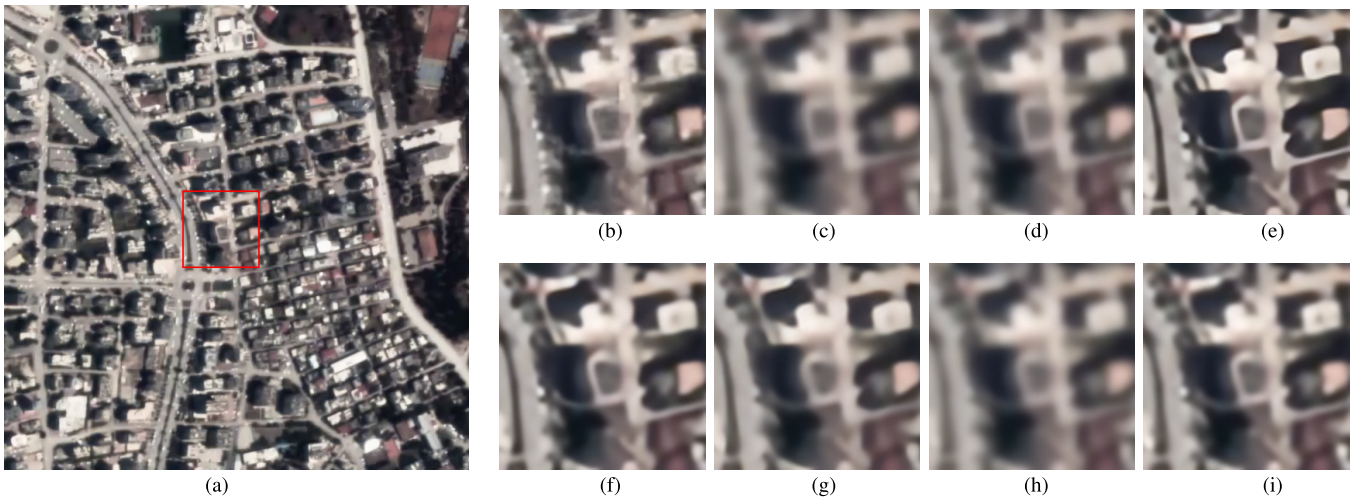


Fig. 10. Qualitative results on the 25th frame of Scene3 with a kernel width of 2.0. The different SR results of the red box area are zoomed in for better visualization. The highest PSNR value is marked in bold. (a) Twenty-fifth frame of Scene3. (b) Ground truth. (c) PVD + MDP (25.47). (d) MDP + PVD (26.40). (e) IKC (26.36). (f) DAN (29.96). (g) DASR (29.15). (h) DBVSR (25.51). (i) Ours (**31.07**).

image. No ground-truth frame is needed to provide in NIQE. In general, the smaller the NIQE, the clearer the texture. For each scene of satellite video clips, NIQE is calculated by averaging over all frames sampled from 11th frame to 20th frame. EDVR [9], DAN [18], and DBVSR [32] are used as representative CNN-based methods for comparison.

The quantitative results are shown in Table III, and the visual results are shown in Figs. 13–16. Considering the value of NIQE and perceptual performance, our proposed JENSVSR obtains better SR results than other comparison methods in most scenes, such as Scene7, Scene8, and Scene10. DAN as a blind image SR method shows better SR performance than EDVR, which is a non-blind video SR method. It is because the degradation is much more complex in real-world satellite videos. It will deteriorate SR performance when the assumed degradation process is not consistent with the real situation. Although DBVSR is designed for blind video SR,

TABLE III

COMPARISON RESULTS OF NIQE ON FOUR SCENES FROM THE JILIN-1 SATELLITE VIDEO TEST DATASET2. FOR EACH SCENE, VALUES OF NIQE ARE AVERAGED OVER FRAMES TAKEN FROM THE 11TH FRAME TO THE 20TH FRAME

	Scene7	Scene8	Scene9	Scene10	Average
EDVR [35]	5.993	6.198	6.093	6.795	6.270
DAN [52]	5.902	6.067	**5.963**	6.672	6.151
DBVSR [60]	6.017	6.493	6.692	6.734	6.484
Ours	**5.656**	**5.988**	6.068	**6.532**	**6.061**

DBVSR seems nonideal to super-resolve satellite videos. DBVSR tendentiously generates over-smooth SR results, where details of texture are over-blurred, e.g., airplanes in Fig. 13, buildings in Fig. 14, and roads in Fig. 14. The phenomenon is basically consistent with synthetic experiments in Section IV-D. Overall, our proposed JENSVSR method has

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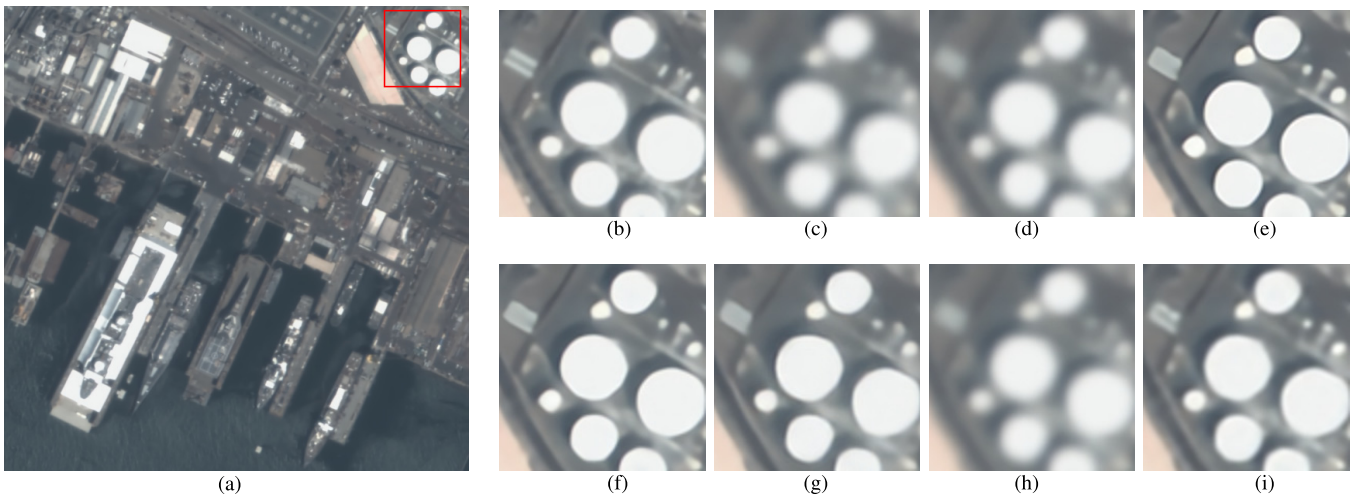


Fig. 11. Qualitative results on the 27th frame of Scene4 with a kernel width of 2.4. The different SR results of the red box area are zoomed in for better visualization. The highest PSNR value is marked in bold. (a) Twenty-seventh frame of Scene4. (b) Ground truth. (c) PVD + MDP (29.95). (d) MDP + PVD (30.65). (e) IKC (33.61). (f) DAN (35.67). (g) DASR (35.42). (h) DBVSR (30.00). (i) Ours (**36.82**).

a better generalization capability to reconstruct satellite videos with clearer details than comparison methods.

In order to further illustrate the influence of SR results on tracking small-sized objects in satellite videos, the tracking results from original LR satellite videos and SR videos by EDVR, DAN, DBVSR, and our proposed JENSVSR, are compared. In terms of small-sized objects in the satellite videos, overlap scores between predicted bounding boxes and ground-truth boxes are more suitable than center location errors. Therefore, the success plot [54] and the area under curve (AUC) of each success plot are used to rank tracking results. The success plot shows the percentage of frames whose overlap scores are larger than given thresholds. The higher the AUC score, the better the tracking performance. Success plots and AUC scores are averaged by all five vehicles with all tracked frames on the Jilin-1 satellite video test dataset2. As shown in Fig. 17, it can be first seen that SR can indeed improve the tracking performance of small-sized vehicles in satellite videos. Second, the relationship between SR performance and tracking performance is a little complex. For example, according to the statistical results of AUC in Fig. 17 and NIQE in Table III, the AUC score of EDVR is higher than DAN, while DAN has better SR results. In addition, over-blurred SR results of DBVSR show lower AUC score than other methods. Finally, our proposed JENSVSR shows the highest AUC score of 57.23%. It means that SR results of our proposed JENSVSR not only have better perceptual SR performance but also are more beneficial to the subsequent tracking task.

F. Ablation Study

1) Effectiveness of the Feature Alignment-Fusion Module: To illustrate the effectiveness of FAFM in SVSRNet, we compare it with three other alignment methods, including alignment by optical flow, alignment by stacking all input frames, and alignment by the pyramid, cascading, deformable convolution (PCD) module in EDVR [9]. These three comparison methods are, respectively, expressed as

TABLE IV

ABLATION STUDY ON THE ALIGNMENT MODULE IN SVSRNET. PSNR AND SSIM ARE CALCULATED BY AVERAGING ON ALL SATELLITE VIDEO CLIPS ON THE JILIN-1 SATELLITE VIDEO TEST DATASET1

Alignment	OF	Stack	PCD	FAFM
PSNR	34.03	34.33	34.19	**35.12**
SSIM	0.9483	0.9497	0.9486	**0.9573**

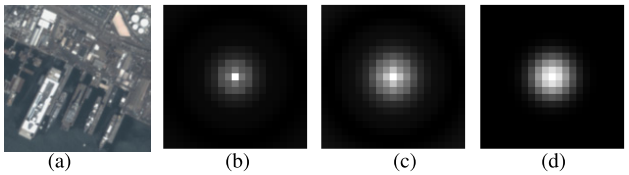


Fig. 12. Visualization of the estimated blur kernels. (a) LR frame. (b) IKC. (c) JENSVSR. (d) Truth map.

SVSRNet-OF, SVSRNet-Stack, and SVSRNet-PCD. Both SVSRNet-OF and SVSRNet-Stack have the same feature extraction and reconstruction modules with SVSRNet. Among them, the only difference is that there is no FAFM in SVSRNet-OF and SVSRNet-Stack. SVSRNet-PCD replaces FAFM with PCD, which applies multi-scale deformable convolutions for alignment. In SVSRNet-OF, adjacent frames are first aligned with the center frame by PWC-Net [55], which is one of the classic optical flow estimation networks. The aligned frames are then input into SVSRNet-OF. The consecutive frames are directly stacked as the inputs of SVSRNet-Stack, which achieves implicit alignment by spatio-temporal convolutions. As shown in Table IV, with the help of FAFM, SVSRNet shows higher PSNR and SSIM values than SVSRNet-OF, SVSRNet-Stack, and SVSRNet-PCD. The visual results are shown in Fig. 18. It can be seen that our proposed SVSRNet can reconstruct more details. Due to accumulated errors caused by optical flow estimation, edges are lost more in SVSRNet-OF than in SVSRNet-Stack. As shown in the research by [56], the phenomenon of training

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(a) (b) (c) (d) (e)

Fig. 13. Real-world experimental results on the 18th frame of Scene7 with an upscale factor of 4. The different SR results of the red box area are zoomed in for better visualization. The best NIQE value is marked in bold. (a) Eighteenth frame of Scene7. (b) EDVR (6.763). (c) DAN (6.666). (d) DBVSR (6.691). (e) Ours (**6.444**).



(a) (b) (c) (d) (e)

Fig. 14. Real-world experimental results on the 15th frame of Scene8 with an upscale factor of 4. The different SR results of the red box area are zoomed in for better visualization. The best NIQE value is marked in bold. (a) Fifteenth frame of Scene8. (b) EDVR (6.189). (c) DAN (6.068). (d) DBVSR (6.500). (e) Ours (**5.987**).



(a) (b) (c) (d) (e)

Fig. 15. Real-world experimental results on the 16th frame of Scene9 with an upscale factor of 4. The different SR results of the red box area are zoomed in for better visualization. The best NIQE value is marked in bold. (a) Sixteenth frame of Scene9. (b) EDVR (6.308). (c) DAN (6.169). (d) DBVSR (6.760). (e) Ours (**6.163**).



(a) (b) (c) (d) (e)

Fig. 16. Real-world experimental results on the 12th frame of Scene10 with an upscale factor of 4. The different SR results of the red box area are zoomed in for better visualization. The best NIQE value is marked in bold. (a) Twelfth frame of Scene10. (b) EDVR (6.072). (c) DAN (5.971). (d) DBVSR (6.058). (e) Ours (**5.699**).

instability in the deformable alignment will deteriorate the SR performance. The SR performance of SVSRNet-PCD is inferior to SVSRNet using FAFM. Both FAFM and PCD achieve alignment in the feature space. PCD implicitly aligns temporal features through its learned offsets. Although learned offsets are highly similar to optical flows, the difference between them cannot be ignored. Therefore, PCD still lacks the clear physical meaning. The core principle of FAFM is to

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TABLE V

ABLATION STUDY ON SIMILARITY MEASURE FUNCTIONS IN SVSRNET. PSNR AND SSIM ARE CALCULATED BY AVERAGING ON ALL SATELLITE VIDEO CLIPS ON THE JILIN-1 SATELLITE VIDEO TEST DATASET1

Method	Euclidean	CC	GJ	GD	Cosine
PSNR	34.92	**35.13**	**35.13**	35.07	35.12
SSIM	0.9556	**0.9574**	0.9573	0.9568	0.9573

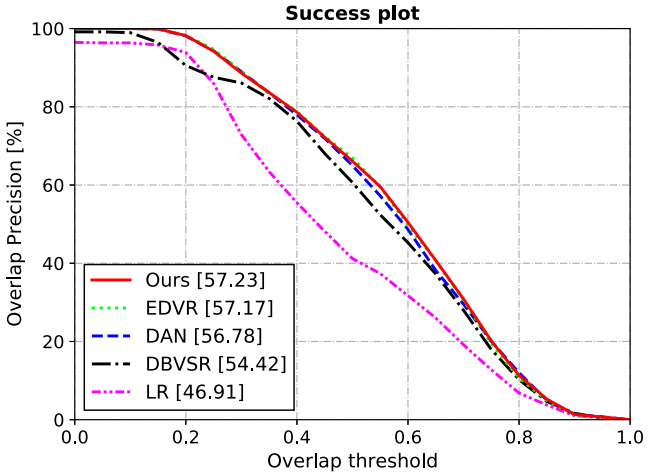


Fig. 17. Averaged success plots and AUC scores of five vehicles on the satellite video test dataset2 based on the SiamRPN++ tracker.

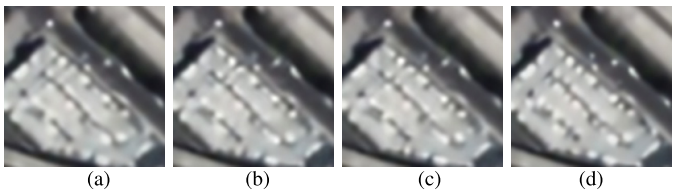


Fig. 18. Visual comparison of different alignment methods for the locally enlarged region on the 15th frame of Scene1. (a) OF. (b) Stack. (c) PCD. (d) FAFM.

achieve patch matching in the feature space, which generates stable and explicit alignment results. Both the quantitative and qualitative results demonstrate that our proposed FAFM can help the SVSRNet generate more satisfactory SR results.

2) Impact of Similarity Measure Function Between Patches: In the FAFM, as shown in (4), cosine coefficients are used to select the most similar patch to the specified patch. To analyze the impact of different similarity measure functions, according to the work in [57], other four similarity measure functions are applied to SVSRNet by replacing cosine coefficients, including the Euclidean distance, correlation coefficients (CC), generalized Jaccard coefficients (GJ), and generalized Dice coefficients (GD). As shown in Table V, the similarity measure function plays an important role in SVSRNet, which directly determines the alignment accuracy. The similarity measure function based on cosine coefficients, CC, GD, and GJ achieves similar SR performance. They are better than the results based on the Euclidean distance. CC, GD, and GJ need extra computation compared with cosine coefficients. Therefore, the default similarity measure function is still set to the function based on cosine coefficients.

TABLE VI

ABLATION STUDY ON NRG AND NRCAB IN JENSVSR. PSNR AND SSIM ARE CALCULATED BY AVERAGING ON ALL SATELLITE VIDEO CLIPS ON THE JILIN-1 SATELLITE VIDEO TEST DATASET1

(nRG, nRCAB)	(5,20)	(15, 20)	(10, 20)	(10, 15)	(10, 25)
PSNR	33.60	**34.07**	33.62	33.59	33.67
SSIM	0.9420	**0.9468**	0.9425	0.9415	0.9431

TABLE VII

PARAMETER NUMBER (M) AND RUNTIME (ms/FRAME) OF DIFFERENT NON-BLIND VSR METHODS

Methods	EDVR [9]	MuCAN [47]	VSRT [48]	BasicVSR [11]	MSDTGP [44]	SVSRNet (ours)
Params	20.7	19.9	43.8	6.3	14.2	16.0
Runtime	827	1701	3820	413	857	877

TABLE VIII

PARAMETER NUMBER (M) AND RUNTIME (ms/FRAME) OF DIFFERENT BLIND SR METHODS

Methods	IKC [17]	DAN [18]	DASR [19]	DBVSR [32]	JENSVSR (ours)	JENSVSR-RG5 (ours)
Params	9.0	4.3	6.0	25.5	21.7	11.2
Runtime	2182	1534	648	1462	1928	1527

3) Impact of the Number of RGs and RCABs: Table VI shows the SR performance in case of different numbers of RGs (nRG) and numbers of RCABs (nRCAB) in our proposed JENSVSR, where the kernel width is 2.0. The default setting of nRG and nRCAB is 10 and 20, respectively, unless special notes. As the nRCAN increases from 5 to 15, PSNR is increased by 0.47 dB. As the nRCAB increases from 15 to 25, it brings an improvement of 0.08 dB. Because nRG is equal to the number of RBs in BKENet, RGs have a greater impact on SR performance than RCABs.

G. Parameter Number and Runtime

In addition to the visual quality and quantitative evaluation, the comparison results on parameter number and runtime are provided in Tables VII and VIII. The runtime per frame is measured when super-resolving the cropped Scene1 video clip of spatial size 144 $\times$ 144 in the same device.

In terms of the non-blind VSR problem with bicubic degradation, as shown in Table VII, BasicVSR spends the shortest runtime per frame with the minimum parameter number compared with other models. Due to the limit of visual transformers, VSRT has the larger model size and longer runtime per frame. Our proposed SVSRNet method has a similar runtime and parameter number to the remaining non-blind VSR methods.

As shown in Table VIII, in IKC, DAN, and DASR, lighter networks are designed for blind image SR. IKC and DAN based on the iterative framework have longer runtime than DASR based on the non-iterative framework. Because of handling with multiple frames at the same time, both DBVSR and our proposed JENSVSR are designed with larger model size than IKC, DAN, and DASR. In our proposed method, when

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the number of RGs is set to 5 (JENSVSR-RG5), it can greatly shorten the model size and speed up runtime. JENSVSR-RG5 can be used as a lightweight alternative to JENSVSR.

H. Limitations

The proposed JENSVSR performs well for SVSR. However, during the training stage, we use (1) as the assumed degradation model for SVSR, where the type of blur kernel $\mathbf{k}$ is also assumed as the isotropic Gaussian blur kernel. When degradations of satellite videos are not followed (1), or the type blur kernel is not consistent with the assumed one, the proposed JENSVSR will not provide satisfactory SR results. There is still a big gap to handle with arbitrary blur kernels for SVSR. In addition, the designed JENSVSR network is still a heavy network. Much lighter networks for SVSR are needed to be researched in the future work.

V. Conclusion

In this article, a deep joint estimation network is presented for satellite videos SR, where SR results and the blur kernel are simultaneously estimated without iterative steps. It can achieve satellite video SR when the blur kernel is unknown. In this article, only the isotropic Gaussian blur kernels are considered. Our proposed method can also be extended to handle with anisotropic Gaussian blur kernels. The proposed network is composed of a video SR subnetwork and a blur kernel estimation subnetwork, where intermediate features from two subnetworks are fused by a cross-task fusion module. This module can enhance the cooperation between two tasks with positive promotion. The experimental results in satellite videos show that the proposed method is efficient to generate SR results with better SR performance and more beneficial representation for the subsequent tracking task. Moreover, the experimental results also show that satellite video SR is one of the possible ways to solve the challenge of tracking small-sized objects in satellite videos.

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